

CHOICE OF EXPERTS IN TRANSFORMER MODELS

Isaac Li & Minh Phan

Department of Computer Science

University of Rochester

Rochester, NY 14627, USA

{ili3,mphan4}@u.rochester.edu

ABSTRACT

As generative AI models grow ever larger in size, the cost of training and inference has grown with them. Sparely activated models, however, have the potential to massively increase the number of trainable parameters while keeping training and inference costs low. A common technique used in such models is Mixture of Experts (MoE), where a model selects different parameters depending on the input. We plan to apply this technique on various parts of transformer models and compare their efficacy.

1 PROPOSAL

There has been much activity in the generative AI sector recently, as models like ChatGPT and Stable Diffusion have received mainstream use and recognition. However, as these models have grown in capability, they have also grown ever larger in size, and the cost of training and inference has grown with them. GPT3, OpenAI’s last-gen model, had 175 billion parameters and took $3.14\text{E}+23$ flops of compute to train (Brown et al., 2020). Their newest model, GPT4, had 1.76 trillion parameters (OpenAI, 2023), a tenfold increase.

1.1 MOTIVATION

These prohibitive costs have led to recent interest in sparely activated models which have the potential to massively increase the quantity of trainable parameters while maintaining training and inference costs. A common technique used in such models is Mixture of Experts (MoE), where a gating function allows the model to select different parameters depending on the input.

1.2 PROBLEM

In the context of transformers, this technique has commonly been applied to the feed-forward block (FFN) within the transformer, as in the Google Switch Transformer (Fedus et al., 2022). However, this technique could also reasonably be applied to the self-attention block (ATTN), or to the transformer as a whole. We plan to apply this technique on these parts and compare their efficacy. We plan to either modify an existing pre-trained Large Language Model (LLM), by allowing one of the transformer blocks to include an MoE layer by duplicating the desired expert, or to create a transformer from scratch applying the mentioned techniques.

1.3 EVALUATION

We then intend to compare these experts, through training cost, inference cost, number of parameters, and accuracy.

1.4 EXPECTED OUTCOME FOR PROGRESS AND FINAL REPORTS

We intend to have at least one working MoE model for the progress report, and to have all of the models and their comparison for the final report.

REFERENCES

- Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners, 2020.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2022.
- OpenAI. Gpt-4 technical report, 2023.